1. Stopping Criteria

In root finding methods, three stopping criteria were implemented:

- Maximum number of iterations: $n \leq n_{max}$
- Accuracy of successive approximations: $|x_n - x_{n-1}| \leq TOLX$
- Functional tolerance:$|f(x_n)| \leq TOLF$

These criteria prevent infinite loops and ensure that the algorithm stops when sufficient accuracy is achieved.

2. Newton Method for Multiple Roots

We applied Newton's method to the function:

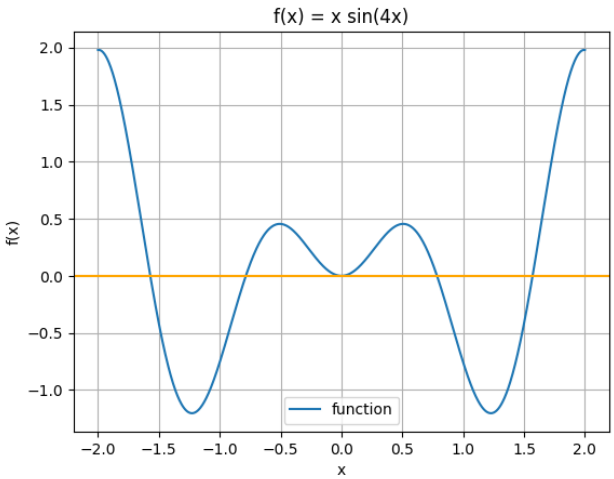
f(x) = x · sin(4x)

This function has multiple roots because:

- x = 0 is a root
- sin(4x) = 0 when 4x = kπ → x = kπ / 4

2.1 Function Plot

The function f(x) = x sin(4x) was plotted to observe its behavior and to verify that it has multiple roots.



2.2 Newton Method – Single Iteration Visualization

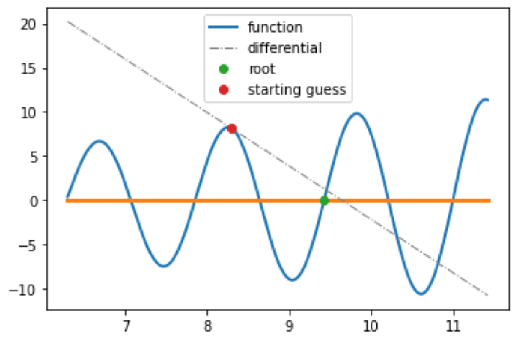
Newton's method uses the following formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

The derivative of the function is:

f'(x) = sin(4x) + 4x cos(4x)

To understand how the method works, one iteration was visualized using the tangent line at an initial guess.



2.3 Results for Different Initial Guesses

Newton's method was applied in a loop for different initial guesses in the interval [-2, 2].

For each initial guess:

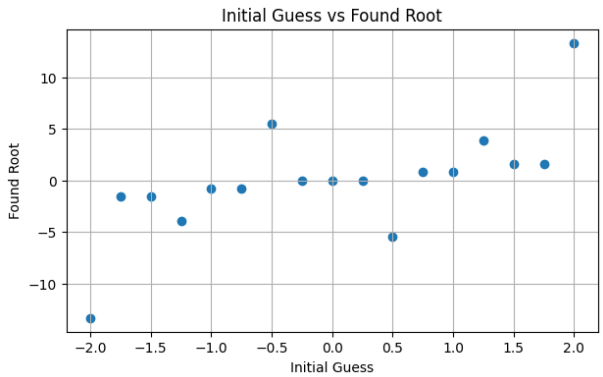
- The root found by the method was recorded
- The number of iterations was recorded

Results Table:

	Initial Guess	Root	Iterations	Converged
0	-2.00	-13.351769	4	True
1	-1.75	-1.570796	4	True
2	-1.50	-1.570796	4	True
3	-1.25	-3.926991	4	True
4	-1.00	-0.785398	4	True
5	-0.75	-0.785398	3	True
6	-0.50	5.497787	4	True
7	-0.25	-0.000006	15	True
8	0.00	0.000000	1	True
9	0.25	0.000006	15	True
10	0.50	-5.497787	4	True
11	0.75	0.785398	3	True
12	1.00	0.785398	4	True
13	1.25	3.926991	4	True
14	1.50	1.570796	4	True
15	1.75	1.570796	4	True
16	2.00	13.351769	4	True

2.4 Initial Guess vs Found Root

The relationship between the initial guess and the root found is shown below.



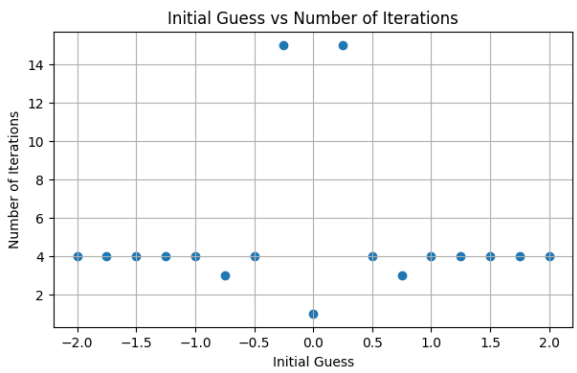
The relationship between initial guess and found root is not continuous.

Small changes in the initial guess may lead to completely different roots.

Therefore, a scatter plot is used instead of a continuous line.

2.5 Initial Guess vs Number of Iterations

The number of iterations required for convergence depending on the initial guess is shown below.



3. Discussion

The results show that Newton's method strongly depends on the initial guess.

- Different initial guesses may converge to different roots.
- In most cases, the method converges in 3–5 iterations.

- However, for some values (e.g. around ±0.25), the number of iterations increases significantly.
- This happens because the derivative f'(x) is close to zero, which causes large jumps in the Newton update formula.

4. Conclusion

Newton's method is a fast and efficient root-finding method. However, its convergence depends on the initial guess. Choosing a good starting point improves both accuracy and convergence speed.